Elecoglipron, an oral small molecule GLP-1 receptor agonist in adults with type 2 diabetes (SOLSTICE): a multicentre, phase 2b, randomised, placebo-controlled trial

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Summary

Background Elecoglipron is an oral, small molecule glucagon-like peptide (GLP)-1 receptor agonist currently in development for the management of type 2 diabetes. Elecoglipron is orally administered once daily with no food or fluid restrictions. SOLSTICE, a phase 2b study, evaluated the efficacy, safety, and tolerability of elecoglipron versus placebo in participants with type 2 diabetes.

Methods This phase 2b, randomised, double-blind, placebo-controlled trial was conducted in medical research centres and hospitals across nine countries, Canada, Germany, Hungary, Japan, Poland, Slovakia, Spain, the UK, and the USA. Sites were selected based on their capacity to fulfil protocol requirement, including, but not limited to, regulatory and ethics approvals, appropriate infrastructure and personnel, and access to the target patient population. Participants aged 18 years and older with a BMI of 23 kg/m² or higher and type 2 diabetes (glycated haemoglobin [HbA_{1c}] ≥7·0% to ≤10·5%, ≥6·5% to ≤10·5% in the USA) managed with diet and exercise alone or monotherapy with metformin or an SGLT2 inhibitor were enrolled. Participants were randomly assigned (3:5:3:5:3:3:6:4) via an interactive web response system to elecoglipron as oral once-daily tablets without dose escalation (5, 15, or 25 mg) or three different dose-escalation regimens, to target doses of 50 mg and 75 mg. The daily dose of 50 mg was evaluated using an every 2-week dose-escalation schedule, while 75 mg was assessed with every 2-week or every 4-week dose-escalation schedules. Participants could also be randomly assigned to placebo matched to each of the elecoglipron groups, or open-label oral semaglutide titrated to 14 mg once daily for 26 weeks. Participants, the treating physician, and the sponsor were masked to doses of elecoglipron or matched placebo, while semaglutide was open-label. The primary endpoint was percent change in HbA_{1c} from baseline to 26 weeks. Efficacy, safety, and tolerability were assessed in all participants who received at least one dose of trial treatment. The trial is registered with ClinicalTrials.gov (NCT06579105) and clinicaltrials.eu (2024-512562-34-00) and is completed.

Findings From Oct 8, 2024, to June 6, 2025, 863 individuals were screened for study inclusion, 457 were excluded as they did not meet the inclusion criteria or met the exclusion criteria, 406 were enrolled and randomly assigned to one of the eight treatment groups, and 404 participants received at least one dose of trial treatment. Among those who received at least one dose of trial treatment, mean (SD) baseline characteristics were: age 58·4 years (10·7); HbA_{1c} of 7·9% (0·9); bodyweight of 99·8 kg (22·1); and BMI of 34·9 kg/m² (7·5). Of 404 participants, 168 (42%) of participants were female and 236 (58%) were male; 280 (69%) were White. At week 26, the mean change from baseline in HbA_{1c} ranged between –0·91% (95% CI –1·25 to –0·58; 5 mg elecoglipron) and –1·88% (–2·23 to –1·53; 75 mg elecoglipron with every 2-week dose-escalation step) compared with –0·15% (–0·42 to 0·12) with placebo. Adverse events were reported by 63% (24 of 38 in the 5 mg group and 39 of 62 in the 15 mg group) to 87% (33 of 38 in 75 mg every 4-week dose escalation) of participants across elecoglipron doses compared with 63% (45 of 71) in the placebo group. The most common adverse events were gastrointestinal, including nausea, constipation, diarrhoea, and vomiting.

Interpretation Once-daily oral elecoglipron showed reductions in glycaemia and a safety and tolerability profile consistent with the GLP-1 receptor agonist class at a similar phase of development, supporting continued development with phase 3 trials for people living with type 2 diabetes.

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Introduction

Type 2 diabetes is a complex, multifactorial metabolic disorder influenced by genetic, behavioural, and environmental risk factors, with a steadily increasing global prevalence in both adult and paediatric populations.[1,2] Type 2 diabetes is also an established independent risk factor for cardiovascular disease and chronic kidney disease, and

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Research in context

Evidence before this study

To inform the design of this phase 2 trial, a structured literature search was conducted in Jan 15, 2024, to identify existing evidence on glucagon-like peptide (GLP)-1 receptor agonists in adults with type 2 diabetes. We searched PubMed, Embase, and ClinicalTrials.gov using terms "type 2 diabetes", "small molecule glucagon-like peptide-1 receptor agonist", "oral glucagon-like peptide-1 receptor agonist", "semaglutide", "orforglipron", "randomised clinical trial", "dose finding", "phase 2", "HbA1c", "glycated haemoglobin", "body weight", "safety", "tolerability", and "adverse events". No date or study duration restrictions were applied; however, non-English references were excluded. Reports in the literature described glycated haemoglobin (Hb) A_{1c} reductions of up to 1·6% with semaglutide, an oral peptide GLP-1 receptor agonist, compared with placebo, and bodyweight reduction of 5·7 kg versus placebo in a phase 2 trial. Adverse events were reported in up to 86% of participants on oral semaglutide. In another phase 2 trial, an oral small molecule GLP-1 receptor agonist, orforglipron showed placebo-adjusted HbA_{1c} reduction of up to 1·67% and placebo-adjusted bodyweight reduction of 7·9 kg. Adverse events were reported in up to 88·9% of participants on oral orforglipron. Due to their efficacy in glycaemic control, weight management, and cardiovascular and renal risk reduction, GLP-1 receptor agonists have an increasingly prominent role in the care of people with type 2 diabetes. Injectable delivery is perceived as a barrier to early adoption of GLP-1 receptor agonists in the treatment of type 2 diabetes and available oral options are limited, with the oral peptide formulation requiring strict fasting restrictions for dose administration to support its bioavailability. Data from a phase 1, first-in-human trial demonstrated that elecoglipron, an oral small molecule GLP-1 receptor agonist, has a safety profile consistent with other GLP-1 receptor agonists and showed dose-dependent reductions in glucose and bodyweight. The phase 2b SOLSTICE trial was designed to further evaluate elecoglipron for the treatment of type 2 diabetes in individuals treated with diet and exercise alone or with metformin or SGLT2 inhibitor monotherapy. In this trial, dose regimens ranging from 5 mg to 75 mg once daily over 26 weeks were evaluated.

Added value of this study

In this phase 2b trial, six dose regimens (from 5 mg to 75 mg) of elecoglipron, an oral small molecule GLP-1 receptor agonist, were compared with placebo to investigate the balance of efficacy and dose-escalation tolerability in participants living with type 2 diabetes. At the highest dose (75 mg) and fastest escalation scheme (every 2-week dose escalation), treatment with elecoglipron resulted in a mean decrease in HbA_{1c} of –1·88% (95% CI –2·23 to –1·53) and mean bodyweight change of –7·7% (–9·4 to –6·1) at week 26. The safety and tolerability profile was consistent with those reported in other GLP-1 receptor agonist trials at a similar phase of development, with adverse events reported by 63% to 87% of participants across elecoglipron groups compared with 63% in the placebo group. Most adverse events were dose-related gastrointestinal side-effects such as nausea, vomiting, diarrhoea, and constipation.

Implications of all the available evidence

In the SOLSTICE trial, treatment with elecoglipron, an oral small molecule GLP-1 receptor agonist, showed improvements in glycaemic control and bodyweight in individuals with type 2 diabetes, with results similar to findings observed in phase 2b type 2 diabetes trials with oral peptide and small molecule GLP-1 receptor agonists. Further, elecoglipron demonstrated an acceptable tolerability and safety profile. Elecoglipron administration requires no food or fluid restrictions and could provide a convenient and accessible treatment option for those living with type 2 diabetes. These trial findings support advancement of elecoglipron to a large phase 3 programme, during which long-term efficacy and safety can be further evaluated in people living with type 2 diabetes.

substantially contributes to morbidity and mortality in adults.[1,3]

In recent years, the class of glucagon-like peptide (GLP)-1 receptor agonists has emerged as an effective therapeutic option for management of type 2 diabetes due to its demonstrated robust glycaemic, weight, cardiovascular, and renal benefits.[4–7] Most of the currently available GLP-1 receptor agonist therapies are administered as subcutaneous injection and the only currently approved oral GLP-1 receptor agonist available for type 2 diabetes—semaglutide—requires administration not only in the fasted state, but also restricts intake of both food and water around the time of dosing. Several oral small molecule GLP-1 receptor agonists are currently in development for the management of type 2 diabetes and obesity.[8,9] One such agent, orforglipron, is now approved in the USA for chronic weight management in adults with obesity, or those who are overweight with at least one weight-related comorbid condition.[10]

Elecoglipron (AZD5004) is an oral small molecule GLP-1 receptor agonist in development for type 2 diabetes and is anticipated to provide similar clinical benefits to other GLP-1 receptor agonists, as well as the potential additional benefits of improved patient access, convenience, and adherence due to its oral administration that does not require fasting or water restrictions.[11] A phase 1, first-in-human trial demonstrated a safety and tolerability profile consistent with other GLP-1 receptor agonists, and dose-dependent reductions in blood glucose and bodyweight.[11]

The phase 2b SOLSTICE trial evaluated the effect of elecoglipron on the lowering of glucose and bodyweight, as well as its safety and tolerability profile versus placebo in participants with inadequately controlled type 2 diabetes on diet and exercise alone, or when treated with a stable dose of metformin or an SGLT2 inhibitor. The trial design also allowed evaluation of multiple different starting doses, dose-escalation schemes, and target maintenance doses regarding both efficacy and tolerability that could inform dose selection for phase 3 development.

Methods

Study design

SOLSTICE was a phase 2b, randomised, double-blind, placebo-controlled trial evaluating once-daily oral elecoglipron versus placebo over 26 weeks in adults with type 2 diabetes. An open-label oral semaglutide group was included for exploratory comparison. The oral semaglutide group was open-label, as encapsulation for masking purposes could delay the release of the active substance. Given its already low bioavailability (about 1%), any delay or disruption might have affected the absorption process and clinical outcomes. The trial was conducted in medical research centres and hospitals across nine countries, Canada, Germany, Hungary, Japan, Poland, Slovakia, Spain, the UK, and the USA. Sites were selected based on their capacity to fulfil protocol requirement, including, but not limited to, regulatory and ethics approvals, appropriate infrastructure and personnel, and access to the target patient population.

Patient insight work was undertaken to support protocol development to minimise the burden of participation and to provide feedback into the informed consent process. The trial was conducted in accordance with the Declaration of Helsinki and Good Clinical Practice guidelines. Local institutional review boards or independent ethics committees at all sites approved the protocol before trial initiation, and all participants provided written informed consent. A listing of ethics review boards and institutional review committees are presented in appendix 1 (p 6). The statistical analysis plan is published in appendix 2. This trial is registered with ClinicalTrials.gov (NCT06579105) and clinicaltrials.eu (2024-512562-34-00) and is completed. Following trial initiation, the glycated haemoglobin (HbA_{1c}) entry criteria in the USA were amended to allow a range of 6·5–10·5%, both to support recruitment efforts and to investigate earlier utilisation of a GLP-1 receptor agonist agent.

See Online for appendix 1

See Online for appendix 2

Participants

Eligible participants were adults aged 18 years and older with type 2 diabetes diagnosed at least 6 months before enrolment. HbA_{1c} at screening was required to be 7·0–10·5% globally (6·5–10·5% in the USA), with a BMI of 23 kg/m² or more. Participants were also required to have type 2 diabetes managed with diet and exercise alone or monotherapy with a stable dose of metformin or a sodium-glucose co-transporter 2 (SGLT2) inhibitor. All background type 2 diabetes management remained unchanged throughout the trial unless a participant initiated hyperglycaemic rescue. Self-reported bodyweight had to be stable (changes only within 5%) for 3 months before randomisation.

Key exclusion criteria included type 1 diabetes, secondary forms of diabetes, and history of diabetic ketoacidosis or hyperosmolar coma or history of proliferative diabetic retinopathy, diabetic maculopathy, or severe non-proliferative diabetic retinopathy. Participants were also excluded if they had clinically

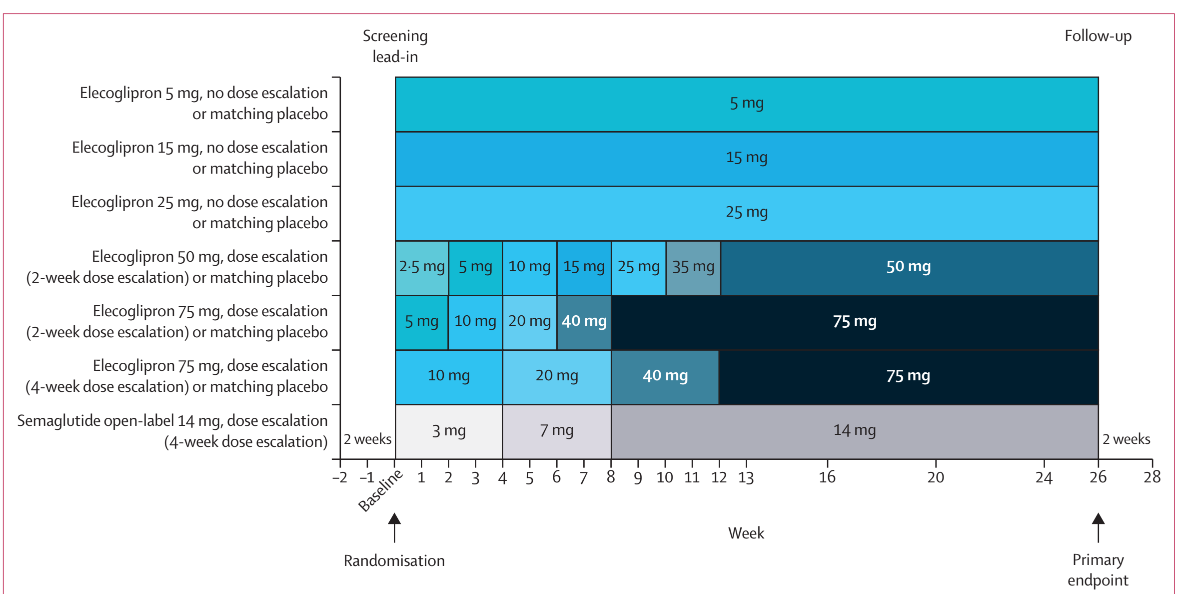

Figure 1: **Dose and dose-escalation schema**

significant gastrointestinal disease affecting gastric emptying or absorption, hepatic impairment, or chronic renal impairment (estimated glomerular filtration rate ≤45 mL/min per 1·73 m²). Full eligibility criteria are presented in appendix 1 (p 7). Sex (male or female), race, and ethnicity were self-reported by participants (options included White, Black or African American, Asian, American Indian or Alaska Native, Native Hawaiian or other Pacific Islander, and other or unknown).

Randomisation and masking

Study site investigators involved in the trial enrolled eligible participants who met full inclusion criteria. The participants were centrally randomised using an interactive web-response computer-based technology randomisation system and trial supply management. Eligible participants were randomly assigned in a 3:5:3:5:3:3:6:4 ratio to receive elecoglipron 5 mg, 15 mg, 25 mg, 50 mg, or 75 mg (two subgroups), matching placebo for each of the six elecoglipron groups, or open-label semaglutide (figure 1). Randomisation was stratified by participants on background SGLT2 inhibitors at baseline; participants not on background SGLT2 inhibitors at baseline and with HbA_{1c} of 8% or less; and participants not on background SGLT2 inhibitors and with HbA_{1c} of more than 8%. Participants, investigators, and sponsor personnel were masked to elecoglipron and placebo assignment; semaglutide was open-label. All doses of trial treatment were administered orally once daily. Elecoglipron and matching placebo tablets were identical in appearance at each dose level.

Procedures

The overall trial duration was up to 30 weeks, including a screening period of up to 1 week, a placebo lead-in period of approximately 1 week, and a treatment duration of 26 weeks. A safety follow-up visit occurred approximately 2 weeks after the last dose of trial intervention.

Elecoglipron and matched placebo were self-administered by participants and dosed in three oral once-daily tablet schedules without dose escalation (5 mg, 15 mg, or 25 mg) and as three different dose-escalation regimens: 50 mg daily dose initiated at 2·5 mg and escalated every 2 weeks and 75 mg daily dose escalated every 2 weeks or every 4 weeks, with starting doses of 5 mg and 10 mg, respectively (figure 1). Oral semaglutide was titrated every 30 days up to a target dose of 14 mg once daily per local prescribing information, taken while fasting, at least 30 min before food or other oral medications, and with up to 120 mL of water.

During the placebo lead-in period, participants received training on the recognition and management of hyperglycaemia and hypoglycaemia, self-monitoring of blood glucose, and diabetes management. During the 26-week intervention period, temporary down-titration of the treatment dose was permitted for gastrointestinal intolerance, with recommendation for re-escalation attempt at the next visit. Treatment was permanently discontinued if participants missed more than 14 consecutive days of trial drug. In case of hyperglycaemia as defined by pre-specified criteria, rescue therapy (ie, treatment intensification or initiation of a new glucose-lowering medication) could be initiated at the investigator's discretion; GLP-1 receptor agonists were not permitted as rescue medication.

Trial visits occurred weekly for weeks 1–4, every 2 weeks for weeks 6–16, then every 4 weeks until the end of treatment; a follow-up visit occurred approximately 14 days after the last dose of trial intervention (figure 1). Efficacy assessments, including glucose and bodyweight were measured at each study visit while fasting glucose (weeks 0, 4, 12, 14, 16, 26) and HbA_{1c} concentrations (weeks 0, 4, 12, 16, 26) were measured at pre-specified timepoints. Participant questionnaires, the Columbia-Suicide Severity Rating Scale (C-SSRS), and Patient Health Questionnaire-9 (PHQ-9) were administered at every study visit while the Diabetes Treatment Satisfaction Questionnaire (DTSQ) was administered at randomisation and weeks 12, 16, and 26. Safety assessments, including physical examination, vital signs, electrocardiograms, and clinical laboratory safety tests were collected at each study visit.

Outcomes

The primary endpoint was change in HbA_{1c} from baseline to week 26. Secondary glycaemic endpoints were change in fasting glucose at weeks 4, 12, 16, and 26; achievement of HbA_{1c} less than 6·5% at week 26; and achievement of HbA_{1c} less than 7·0% at week 26 in participants with baseline HbA_{1c} of 7·0% or more. Secondary endpoints were absolute and percent change in bodyweight from baseline to week 26, and participants who achieved weight reductions of at least 5%, at least 10%, and at least 15% at week 26.

Harms were assessed systematically throughout the study. Safety and tolerability were assessed, including the occurrence of adverse events, serious adverse events, adverse events leading to treatment discontinuation or dose reduction, and adverse events possibly related to trial intervention as assessed by the investigator. Vital signs, electrocardiogram (ECG), and centrally assessed laboratory parameters were also evaluated.

Statistical analysis

A planned sample size of approximately 384 participants randomly assigned to one of eight treatment groups provided greater than 90% power to detect a difference of 0·9% in HbA_{1c} between any elecoglipron dose level and placebo, assuming a standard deviation of 1·1% at a two-sided α of 0·05. The type 1 error was not controlled for multiplicity and p values should be interpreted descriptively.

Efficacy and safety analyses were conducted in the full analysis set, defined as all randomly assigned participants who received at least one dose of trial drug and analysed according to randomised assignment. Placebo groups were pooled for analysis. Elecoglipron groups with the same target dose were pooled regardless of dose-escalation schedule, unless otherwise specified.

The efficacy estimand (used in the main analysis of primary, secondary, and exploratory endpoints) included only data collected before any intercurrent event (first of permanent discontinuation of study intervention, rescue medication, or death), thereby estimating the treatment effect as if any intercurrent event did not happen. In the efficacy estimand for all endpoints involving bodyweight, the bariatric surgery intercurrent event replaced the rescue medication intercurrent event. A supplementary estimand including data after permanent discontinuation or rescue medication was performed to assess the treatment benefit under a different handling of intercurrent events, thereby enhancing robustness of the efficacy estimand.

The change from baseline in HbA_{1c} was analysed using a mixed model for repeated measures comprising the fixed effects of baseline HbA_{1c}, randomisation stratum, treatment group, visit and interaction of visit, and treatment group. A similar analysis was performed for all continuous efficacy endpoints.

The proportion of participants achieving HbA_{1c} less than 6·5% at week 26 was analysed using a logistic regression model with baseline HbA_{1c}, randomisation

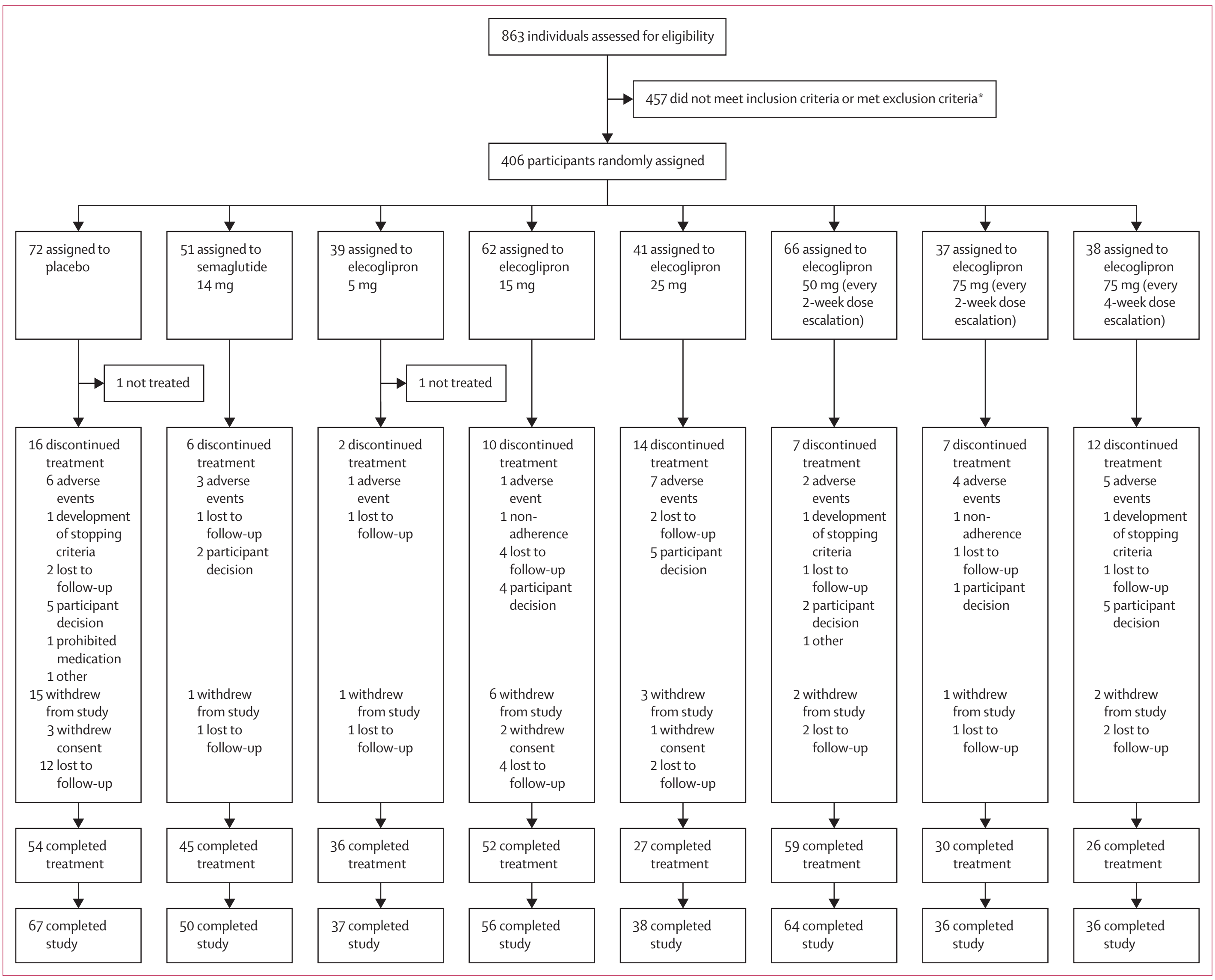

Figure 2: **Trial profile**

Discontinued treatment categorisation reflects primary reason for discontinuation as assessed by investigator. HbA_{1c}=glycated haemoglobin. *The three most common reasons for screen failure were: inclusion criteria not met for HbA_{1c} value at screening of ≥7·0% and ≤10·5% with diet and exercise alone or with a stable dose of metformin or SGLT2 inhibitor for at least 1 month before screening (n=168); exclusion for being on prohibited background medications (n=51); exclusion for presence of hepatitis B core antibody or hepatitis surface B antigen (n=33).

stratum, and treatment group as covariates. Missing data was handled using multiple imputation under missing at random assumptions. A similar analysis was performed for all binary efficacy endpoints.

Safety outcomes were assessed in the full analysis set, including all randomly assigned participants who received at least one dose of study intervention, and according to the treatment groups that participants were randomly assigned. The full statistical analyses plan is presented in appendix 2.

Role of the funding source

The funder of the trial contributed to study design, study conduct, data collection, and data analyses, data interpretation, and writing of the report.

Results

Between Oct 8, 2024, and June 6, 2025, 863 participants were screened, 457 did not meet the inclusion criteria or met the exclusion criteria, 406 were enrolled and randomly assigned to one of eight treatment groups, and 404 received at least one dose of trial treatment (figure 2). In total, 384 (95%) participants completed the trial, and 329 (81%) completed 26 weeks of treatment. The efficacy and safety population included the 404 randomly assigned participants who received at least one dose of trial intervention. In total, 74 (18%) participants discontinued elecoglipron, placebo, or semaglutide. Primary reasons for treatment discontinuation according to the investigators were: 29 (39%) due to adverse events, 24 (32%) due to participant decision, and 13 (18%) were lost to follow-up (appendix 1 p 13).

Summary of participant baseline demographics and clinical characteristics are presented in table 1. Among those who received at least one dose of trial treatment, trial participants had a mean age of 58·4 years (SD 10·7), baseline HbA_{1c} of 7·9% (0·9; 63 mmol/mol), baseline BMI of 34·9 kg/m² (22·1), and median duration of type 2

	Placebo (n=71)	Elecoglipron						Semaglutide 14 mg (every 4-week dose escalation; n=51)
		5 mg (n=38)	15 mg (n=62)	25 mg (n=41)	50 mg (every 2-week dose escalation; n=66)	75 mg (every 2-week dose escalation; n=37)	75 mg (every 4-week dose escalation; n=38)	
Demographic variables								
Sex								
Female	22 (31%)	18 (47%)	29 (47%)	17 (41%)	37 (56%)	10 (27%)	11 (29%)	24 (47%)
Male	49 (69%)	20 (53%)	33 (53%)	24 (59%)	29 (44%)	27 (73%)	27 (71%)	27 (53%)
Age, years	60·1 (10·6)	59·1 (9·6)	56·0 (10·5)	56·8 (12·0)	57·7 (10·2)	58·9 (11·0)	58·6 (10·9)	59·8 (11·1)
Race								
White	47 (66%)	29 (76%)	45 (73%)	33 (80%)	41 (62%)	29 (78%)	17 (45%)	39 (76%)
Black or African American	8 (11%)	4 (11%)	7 (11%)	4 (10%)	16 (24%)	2 (5%)	9 (24%)	10 (20%)
Asian	12 (17%)	5 (13%)	8 (13%)	3 (7%)	7 (11%)	4 (11%)	10 (26%)	2 (4%)
American Indian or Alaska Native	1 (1%)	0	0	0	0	1 (3%)	0	0
Other or unknown	3 (4%)	0	2 (3%)	1 (2%)	2 (3%)	1 (3%)	2 (5%)	0
Ethnicity								
Hispanic or Latino	11 (15%)	3 (8%)	8 (13%)	8 (20%)	6 (9%)	5 (14%)	0	10 (20%)
Not Hispanic or Latino	60 (85%)	35 (92%)	52 (84%)	33 (80%)	60 (91%)	32 (86%)	36 (95%)	39 (76%)
Not reported	0	0	2 (3%)	0	0	0	2 (5%)	2 (4%)
Clinical variables								
HbA_{1c}, %	8·0% (0·8)	7·9% (1·0)	7·9% (1·0)	7·9% (1·0)	7·9% (0·9)	8·1% (1·1)	7·8% (1·0)	7·8% (0·9)
Diabetes duration, years	6·2 (3·2–9·9)	4·8 (2·7–10·4)	6·6 (2·8–10·4)	6·0 (2·7–10·2)	5·3 (2·3–9·9)	7·3 (3·5–13·0)	6·3 (3·3–10·3)	5·9 (1·6–8·3)
Diabetes medication at baseline	52 (73%)	27 (71%)	46 (74%)	34 (83%)	50 (76%)	31 (84%)	34 (89%)	41 (80%)
Bodyweight, kg	100·91 (23·24)	95·74 (22·67)	101·30 (23·49)	107·12 (20·25)	99·85 (23·87)	98·53 (20·40)	96·43 (20·94)	97·35 (18·68)
BMI, kg/m²	34·7 (7·2)	33·7 (6·5)	35·8 (8·5)	37·6 (7·5)	35·8 (9·0)	33·0 (6·5)	32·7 (7·0)	34·4 (5·6)
eGFR, mL/min per 1·73 m²	93·7 (15·3)	92·3 (19·2)	97·6 (14·8)	95·6 (15·8)	91·3 (16·5)	92·8 (14·0)	95·1 (16·8)	89·2 (16·4)
Systolic blood pressure, mm Hg	127·9 (15·9)	129·3 (11·8)	127·5 (13·8)	130·5 (16·4)	130·6 (13·7)	130·7 (12·1)	131·9 (16·6)	129·8 (10·4)

Data are mean (SD), n (%), or median (IQR). Starting dose for elecoglipron 50 mg every 2-week dose escalation: 2·5 mg; for elecoglipron 75 mg every 2-week dose escalation: 5 mg; for elecoglipron 75 mg every 4-week dose escalation: 10 mg; for semaglutide every 4-week dose escalation: 3 mg. Percentages may not add to total 100 because of rounding. eGFR=estimated glomerular filtration rate. HbA_{1c}=glycated haemoglobin.

Table 1: **Baseline characteristics**

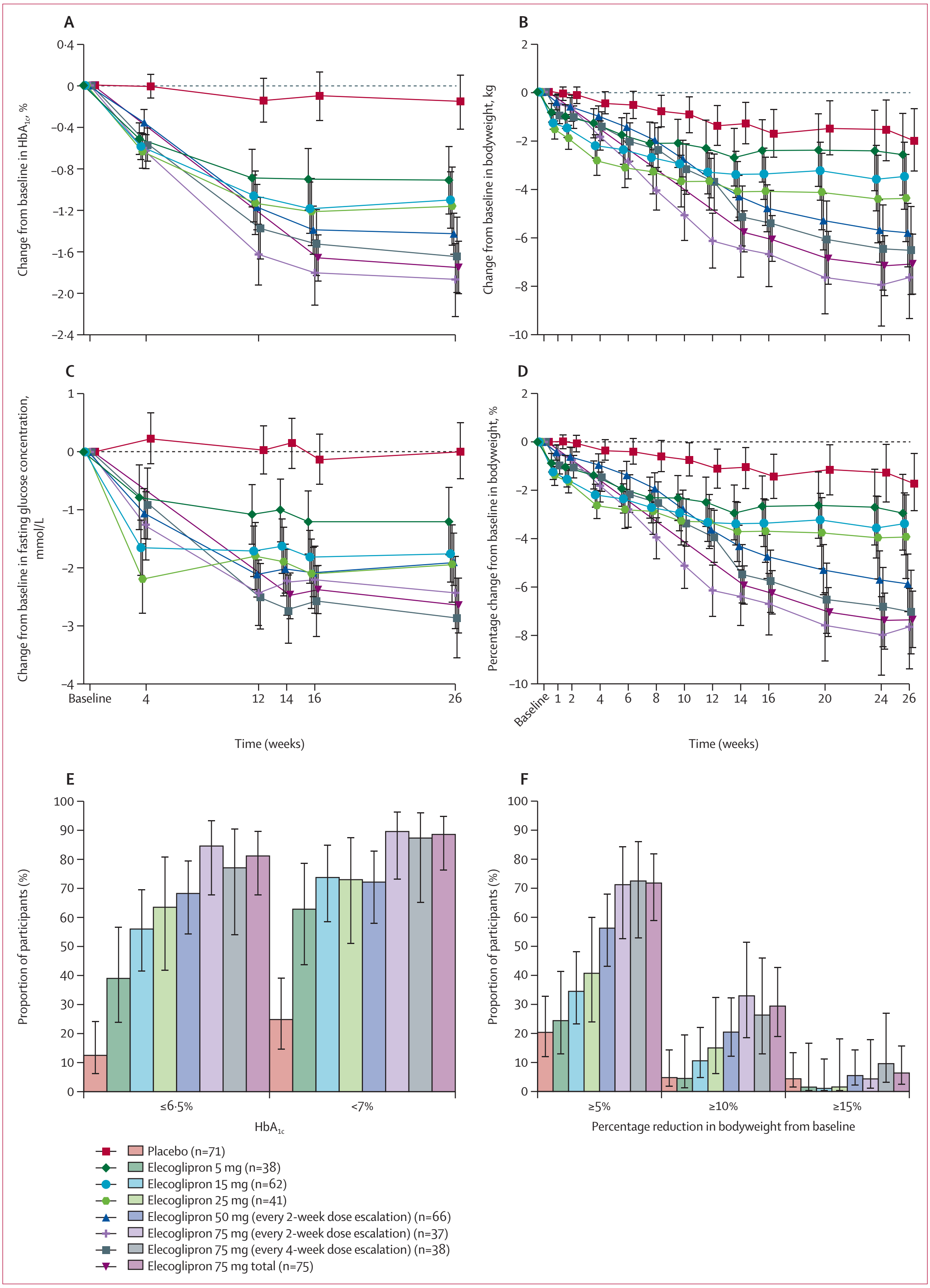

***Figure 3*: Efficacy parameters**
(A) Change from baseline in HbA_{1c}, %. (B) Change from baseline in bodyweight (kg). (C) Change from baseline in fasting glucose concentration (mmol/L). (D) Percentage change from baseline in bodyweight. (E) Proportion of participants reaching HbA_{1c} ≤6·5% and <7% at week 26. (F) Proportion of participants reaching at least 5%, at least 10%, and at least 15% reduction in bodyweight from baseline at week 26. HbA_{1c}=glycated haemoglobin.

	Placebo (n=71)	Elecoglipron							Semaglutide 14 mg (every 4-week dose escalation; n=51)
		5 mg (n=38)	15 mg (n=62)	25 mg (n=41)	50 mg (every 2-week dose escalation; n=66)	75 mg (every 2-week dose escalation; n=37)	75 mg (every 4-week dose escalation; n=38)	75 mg pooled (n=75)	
HbA_{1c}, %									
Baseline	7·96 (0·83)	7·93 (0·95)	7·93 (0·95)	7·92 (0·95)	7·88 (0·94)	8·08 (1·06)	7·78 (1·00)	7·93 (1·03)	7·82 (0·88)
Change from baseline at 26 weeks (95% CI)	−0·15 (−0·42 to 0·12)	−0·91 (−1·25 to −0·58)	−1·11 (−1·38 to −0·83)	−1·16 (−1·54 to −0·79)	−1·42 (−1·68 to −1·16)	−1·88 (−2·23 to −1·53)	−1·64 (−2·00 to −1·27)	−1·76 (−2·01 to −1·50)	−1·28 (−1·58 to −0·98)
Compared with placebo	..	−0·76 (−1·19 to −0·34); p=0·0005	−0·96 (−1·34 to −0·57); p<0·0001	−1·01 (−1·47 to −0·55); p<0·0001	−1·27 (−1·64 to −0·90); p<0·0001	−1·73 (−2·17 to −1·29); p<0·0001	−1·49 (−1·94 to −1·03); p<0·0001	−1·61 (−1·97 to −1·24); p<0·0001	..
Compared with semaglutide	..	0·37 (−0·08 to 0·82)	0·18 (−0·23 to 0·59)	0·12 (−0·36 to 0·60)	−0·14 (−0·54 to 0·26)	−0·60 (−1·06 to −0·13)	−0·35 (−0·83 to 0·12)	−0·47 (−0·87 to −0·08)	..
Fasting serum glucose, mmol/L									
Baseline	8·65 (1·87)	8·98 (2·18)	8·72 (2·38)	8·74 (2·24)	8·93 (2·73)	9·47 (2·42)	8·43 (2·52)	8·96 (2·51)	8·61 (1·94)
Change from baseline at 26 weeks (95% CI)	0·02 (−0·47 to 0·51)	−1·19 (−1·78 to −0·60)	−1·74 (−2·24 to −1·24)	−1·92 (−2·60 to −1·24)	−1·85 (−2·31 to −1·39)	−2·41 (−3·03 to −1·79)	−2·86 (−3·54 to −2·18)	−2·64 (−3·10 to −2·18)	−1·68 (−2·22 to −1·14)
Compared with placebo	..	−1·21 (−1·98 to −0·45); p=0·0020	−1·76 (−2·46 to −1·06); p<0·0001	−1·94 (−2·78 to −1·11); p<0·0001	−1·87 (−2·54 to −1·20); p<0·0001	−2·44 (−3·23 to −1·64); p<0·0001	−2·88 (−3·72 to −2·04); p<0·0001	−2·66 (−3·33 to −1·99) p<0·0001	..
Compared with semaglutide	..	0·49 (−0·31 to 1·29)	−0·06 (−0·79 to 0·67)	−0·24 (−1·10 to 0·63)	−0·16 (−0·87 to 0·54)	−0·73 (−1·55 to 0·09)	−1·18 (−2·04 to −0·31)	−0·96 (−1·67 to −0·25)	..
Fasting serum glucose, percentage change									
Percentage change from baseline at 26 weeks (95% CI)	2·5 (−3·3 to 8·3)	−12·6 (−19·5 to −5·6)	−17·5 (−23·3 to −11·6)	−20·7 (−28·7 to −12·8)	−16·8 (−22·2 to −11·3)	−24·7 (−32·1 to −17·4)	−29·4 (−37·4 to −21·3)	−27·1 (−32·5 to −21·7)	−17·9 (−24·3 to −11·6)
Compared with placebo	..	−15·1 (−24·1 to −6·0); p=0·0012	−20·0 (−28·2 to −11·8); p<0·0001	−23·2 (−33·1 to −13·4); p<0·0001	−19·3 (−27·2 to −11·4); p<0·0001	−27·3 (−36·6 to −17·9); p<0·0001	−31·9 (−41·8 to −22·0); p<0·0001	−29·6 (−37·5 to −21·7); p<0·0001	..
Compared with semaglutide	..	5·4 (−4·1 to 14·8)	0·4 (−8·2 to 9·1)	−2·8 (−13·0 to 7·4)	1·1 (−7·2 to 9·5)	−6·8 (−16·5 to 2·9)	−11·4 (−21·7 to −1·2)	−9·2 (−17·5 to −0·8)	..
Bodyweight, kg									
Baseline	100·91 (23·24)	95·74 (22·67)	101·30 (23·49)	107·12 (20·25)	99·85 (23·87)	98·53 (20·40)	96·43 (20·94)	97·46 (20·56)	97·35 (18·68)
Change from baseline at 26 weeks (95% CI)	−1·94 (−3·25 to −0·64)	−2·54 (−4·25 to −0·83)	−3·42 (−4·81 to −2·03)	−4·35 (−6·19 to −2·50)	−5·79 (−7·09 to −4·48)	−7·61 (−9·39 to −5·84)	−6·52 (−8·36 to −4·68)	−7·06 (−8·34 to −5·78)	−4·69 (−6·19 to −3·18)
Compared with placebo	..	−0·60 (−2·75 to 1·55); p=0·59	−1·48 (−3·39 to 0·43); p=0·13	−2·40 (−4·67 to −0·14); p=0·038	−3·84 (−5·69 to −2·00); p<0·0001	−5·67 (−7·87 to −3·47); p<0·0001	−4·58 (−6·83 to −2·32); p<0·0001	−5·11 (−6·94 to −3·29); p<0·0001	..
Compared with semaglutide	..	2·15 (−0·13 to 4·42)	1·27 (−0·78 to 3·32)	0·34 (−2·04 to 2·73)	−1·10 (−3·09 to 0·89)	−2·92 (−5·25 to −0·60)	−1·83 (−4·20 to 0·54)	−2·37 (−4·34 to −0·40)	..
Bodyweight, percentage change									
% change from baseline at 26 weeks (95% CI)	−1·7 (−2·9 to −0·5)	−2·9 (−4·5 to −1·3)	−3·4 (−4·7 to −2·1)	−4·0 (−5·7 to −2·2)	−5·9 (−7·1 to −4·6)	−7·7 (−9·4 to −6·1)	−7·0 (−8·8 to −5·3)	−7·4 (−8·6 to −6·2)	−5·1 (−6·5 to −3·7)
Compared with placebo	..	−1·3 (−3·3 to 0·8); p=0·22	−1·7 (−3·5 to 0·1); p=0·064	−2·3 (−4·4 to −0·1); p=0·038	−4·2 (−5·9 to −2·4); p<0·0001	−6·1 (−8·1 to −4·0); p<0·0001	−5·3 (−7·5 to −3·2); p<0·0001	−5·7 (−7·4 to −4·0); p<0·0001	..
Compared with semaglutide	..	2·1 (0·0 to 4·3)	1·7 (−0·2 to 3·6)	1·1 (−1·1 to 3·4)	−0·8 (−2·7 to 1·1)	−2·7 (−4·9 to −0·5)	−2·0 (−4·2 to 0·3)	−2·3 (−4·2 to −0·5)	..

Data are mean (SD) or least squares mean differences (95% CI) and p values from mixed models for repeated measures. Starting dose for elecoglipron 50 mg every 2-week dose escalation: 2·5 mg; for elecoglipron 75 mg every 2-week dose escalation: 5 mg; for elecoglipron 75 mg every 4-week dose escalation: 10 mg; for semaglutide every 4-week dose escalation: 3 mg. HbA_{1c}=glycated haemoglobin.

Table 2: **Primary and key secondary efficacy endpoints at week 26**

diabetes of 6·2 years (IQR 2·7–10·2). Of 404 participants who received at least one dose of trial treatment, there were 168 (42%) females and 236 (58%) males. A slight sex imbalance was noted in the elecoglipron 75 mg treatment groups (27–29% female), the placebo group (31% female), and the elecoglipron group 50 mg treatment group (56%) compared with the other treatment groups which included 42–47% females (table 1). Most participants were White (280 [69%]), followed by Black or African American (60 [15%]), and Asian (51 [13%]). Baseline demographics and clinical parameters were otherwise similar across the eight treatment groups. Rescue therapy for hyperglycaemia was required by 3·2% of elecoglipron participants (two participants in the 5 mg group, two participants in the 15 mg group, one participant in the 25 mg group, two participants in the 50 mg group, and two participants in the 75 mg every 2-week dose-escalation group) compared with 12·7% of participants in the placebo group (appendix 1 p 13).

Greater reductions in HbA$_{1c}$ were observed with higher doses of elecoglipron at 26 weeks (figure 3). At

	Elecoglipron							**Semaglutide 14 mg (every 4-week dose escalation; n=51)**	**Placebo (n=71)**
	5 mg (n=38)	15 mg (n=62)	25 mg (n=41)	50 mg (every 2-week dose escalation; n=66)	75 mg (every 2-week dose escalation; n=37)	75 mg (every 4-week dose escalation; n=38)	Total (n=282)		
Overall summary of adverse events									
Any adverse events	24 (63%)	39 (63%)	31 (76%)	47 (71%)	25 (68%)	33 (87%)	199 (71%)	35 (69%)	45 (63%)
Any serious adverse events	0	5 (8%)	4 (10%)	3 (5%)	2 (5%)	0	14 (5%)	1 (2%)	6 (8%)
Any serious adverse event with outcome of death	0	0	0	0	0	0	0	0	1 (1%)
Any adverse event leading to discontinuation of trial treatment	1 (3%)	1 (2%)	7 (17%)	3 (5%)	4 (11%)	6 (16%)	22 (8%)	3 (6%)	6 (8%)
Any adverse event leading to interruption of trial treatment	3 (8%)	5 (8%)	2 (5%)	5 (8%)	3 (8%)	3 (8%)	21 (7%)	4 (8%)	5 (7%)
Any adverse event leading to reduction of trial treatment	0	0	0	4 (6%)	5 (14%)	6 (16%)	15 (5%)	0	1 (1%)
Adverse events possibly related to trial study treatment*	12 (32%)	17 (27%)	26 (63%)	31 (47%)	20 (54%)	24 (63%)	130 (46%)	17 (33%)	18 (25%)
Any gastrointestinal adverse event leading to discontinuation of trial treatment	1 (3%)	0	5 (12%)	1 (2%)	1 (3%)	2 (5%)	10 (4%)	3 (6%)	2 (3%)
Gastrointestinal event by maximum severity (CTCAE grade)									
Any grade	12 (32%)	23 (37%)	26 (63%)	31 (47%)	17 (46%)	27 (71%)	136 (48%)	16 (31%)	19 (27%)
Grade ≥3 (severe or higher)	0	0	1 (2%)†	0	0	0	1 (<1%)†	0	0
Adverse events of special interest									
Hypoglycaemia (potential risk)	0	1 (2%)	0	1 (2%)	1 (3%)	0	3 (1%)	0	0
Blood glucose decreased	0	0	0	1 (2%)	0	0	1 (<1%)	0	0
Hypoglycaemia	0	1 (2%)	0	0	1 (3%)	0	2 (1%)	0	0
Gallbladder-related disorders (potential risk)	0	0	0	1 (2%)	1 (3%)	0	2 (1%)	1 (2%)	2 (3%)
Cholelithiasis	0	0	0	1 (2%)	1 (3%)	0	2 (1%)	1 (2%)	1 (1%)
Acute cholecystitis necrotic	0	0	0	0	0	0	0	0	1 (1%)
Renal impairment (potential risk)	0	0	0	2 (3%)	0	1 (3%)	3 (1%)	2 (4%)	3 (4%)
Blood creatinine increased	0	0	0	1 (2%)	0	0	1 (<1%)	0	0
Proteinuria	0	0	0	0	0	0	0	1 (2%)	0
Renal impairment	0	0	0	0	0	0	0	0	2 (3%)
Chronic kidney disease	0	0	0	0	0	0	0	0	1 (1%)
Acute kidney injury	0	0	0	1 (2%)	0	1 (3%)	2 (1%)	1 (2%)	0
Medullary thyroid cancer (potential risk)	0	0	0	0	0	0	0	0	0
Acute pancreatitis (narrow)	0	0	0	0	0	0	0	0	0

(Table 3 continues on next page)

	Elecoglipron							Semaglutide 14 mg (every 4-week dose escalation; n=51)	Placebo (n=71)
	5 mg (n=38)	15 mg (n=62)	25 mg (n=41)	50 mg (every 2-week dose escalation; n=66)	75 mg (every 2-week dose escalation; n=37)	75 mg (every 4-week dose escalation; n=38)	Total (n=282)		
(Continued from previous page)									
Gastrointestinal adverse events occurring in at least 5% of participants in any group									
Any gastrointestinal adverse events	12 (32%)	23 (37%)	26 (63%)	31 (47%)	17 (46%)	27 (71%)	136 (48%)	16 (31%)	19 (27%)
Nausea	1 (3%)	10 (16%)	11 (27%)	14 (21%)	9 (24%)	14 (37%)	59 (21%)	9 (18%)	2 (3%)
Constipation	5 (13%)	6 (10%)	8 (20%)	11 (17%)	3 (8%)	11 (29%)	44 (16%)	3 (6%)	3 (4%)
Diarrhoea	3 (8%)	5 (8%)	7 (17%)	9 (14%)	2 (5%)	8 (21%)	34 (12%)	1 (2%)	11 (15%)
Vomiting	1 (3%)	6 (10%)	6 (15%)	5 (8%)	4 (11%)	7 (18%)	29 (10%)	2 (4%)	1 (1%)
Dyspepsia	2 (5%)	1 (2%)	3 (7%)	2 (3%)	6 (16%)	3 (8%)	17 (6%)	1 (2%)	2 (3%)
Abdominal pain (upper)	1 (3%)	4 (6%)	1 (2%)	1 (2%)	2 (5%)	3 (8%)	12 (4%)	0	1 (1%)
Eructation	2 (5%)	3 (5%)	2 (5%)	4 (6%)	0	1 (3%)	12 (4%)	1 (2%)	3 (4%)
Flatulence	2 (5%)	2 (3%)	3 (7%)	3 (5%)	0	1 (3%)	11 (4%)	0	2 (3%)
Gastro-oesophageal reflux disease	2 (5%)	1 (2%)	1 (2%)	2 (3%)	0	2 (5%)	8 (3%)	1 (2%)	0
Abdominal pain	0	0	0	0	1 (3%)	2 (5%)	3 (1%)	1 (2%)	1 (1%)
Haemorrhoids	2 (5%)	0	1 (2%)	0	0	0	3 (1%)	0	0
Change of bowel habits	0	0	0	0	0	2 (5%)	2 (1%)	0	0
Adverse events occurring in at least 5% of participants in any group (non-gastrointestinal)									
Decreased appetite	3 (8%)	1 (2%)	5 (12%)	3 (5%)	2 (5%)	2 (5%)	16 (6%)	1 (2%)	3 (4%)
Headache	1 (3%)	2 (3%)	3 (7%)	4 (6%)	1 (3%)	3 (8%)	14 (5%)	0	4 (6%)
Lipase increased	3 (8%)	1 (2%)	0	4 (6%)	4 (11%)	2 (5%)	14 (5%)	0	0
Dizziness	3 (8%)	1 (2%)	2 (5%)	2 (3%)	3 (8%)	2 (5%)	13 (5%)	0	8 (11%)
Upper respiratory tract infection	2 (5%)	4 (6%)	2 (5%)	1 (2%)	0	2 (5%)	11 (4%)	3 (6%)	5 (7%)
Arthralgia	1 (3%)	1 (2%)	2 (5%)	1 (2%)	1 (3%)	4 (11%)	10 (4%)	2 (4%)	4 (6%)
Fatigue	2 (5%)	1 (2%)	2 (5%)	1 (2%)	1 (3%)	3 (8%)	10 (4%)	2 (4%)	4 (6%)
Amylase increased	1 (3%)	1 (2%)	0	2 (3%)	2 (5%)	2 (5%)	8 (3%)	0	0
Nasopharyngitis	1 (3%)	2 (3%)	2 (5%)	1 (2%)	1 (3%)	0	7 (2%)	1 (2%)	4 (6%)
Urinary tract infection	1 (3%)	1 (2%)	1 (2%)	1 (2%)	1 (3%)	2 (5%)	7 (2%)	2 (4%)	1 (1%)
Hypertension	1 (3%)	3 (5%)	0	0	0	2 (5%)	6 (2%)	2 (4%)	4 (6%)
Influenza	0	1 (2%)	0	1 (2%)	2 (5%)	1 (3%)	5 (2%)	0	0
Increased appetite	2 (5%)	0	1 (2%)	0	0	0	3 (1%)	0	0
Insomnia	0	0	1 (2%)	0	0	2 (5%)	3 (1%)	0	0
Rash	0	0	0	0	2 (5%)	1 (3%)	3 (1%)	2 (4%)	0
Gout	0	0	0	0	0	2 (5%)	2 (1%)	0	2 (3%)
Hyperglycaemia	0	0	1 (2%)	1 (2%)	0	0	2 (1%)	1 (2%)	7 (10%)
Neck pain	0	0	0	0	0	2 (5%)	2 (1%)	0	0
Thirst	2 (5%)	0	0	0	0	0	2 (1%)	0	0
Viral infection	2 (5%)	0	0	0	0	0	2 (1%)	0	0

Data are n (%). The table includes adverse events with an onset date on or after the day of the randomisation visit and participants were censored at the earliest of withdrawal of consent, death, or the date of the last clinical assessment. Only the first event per participant in each category is used in the analyses. Starting dose for elecoglipron 50 mg every 2-week dose escalation: 2·5 mg; for elecoglipron 75 mg every 2-week dose escalation: 5 mg; for elecoglipron 75 mg every 4-week dose escalation: 10 mg; for semaglutide every 4-week dose escalation: 3 mg. Data in this table reflect actions taken with study drug for each adverse event while data in figure 2 and appendix 1 (p 13) reflect primary reason for early discontinuation of study drug as assessed by investigator. CTCAE=Common Terminology Criteria for Adverse Events. *Possibly related was defined as a reasonable possibility that the adverse event was caused by trial treatment, as assessed by investigator. †Adverse events of large intestinal haemorrhage.

Table 3: **Adverse events**

week 26, mean change in HbA_{1c} from baseline for elecoglipron was –0·91% (95% CI –1·25 to –0·58) for 5 mg, –1·11% (–1·38 to –0·83) for 15 mg, –1·16% (–1·54 to –0·79) for 25 mg, –1·42% (–1·68 to –1·16) for 50 mg, –1·88% (–2·23 to –1·53) for 75 mg (every 2-week escalation), –1·64% (–2·00 to –1·27) for 75 mg (every 4-week escalation), –0·15% (–0·42 to 0·12) for placebo, and –1·28% (–1·58 to –0·98) for oral semaglutide. All doses of elecoglipron were superior to placebo in lowering HbA_{1c} (p=0·0005 for 5 mg group and p<0·0001 for all other groups) with an estimated treatment difference of –0·76% to –1·73% (table 2).

In the responder analyses, 62·9% of participants in the 5 mg group, 73·8% in the 15 mg group, 73·0% in the 25 mg group, 72·2% in the 50 mg group, 89·6% in the 75 mg (every 2-week escalation) group, and 87·4% in the 75 mg (every 4-week escalation) group with baseline HbA_{1c} greater than or equal to 7·0% who received elecoglipron achieved an HbA_{1c} of less than 7·0% at week 26, versus 24·9% in the placebo group, and 73·6% for oral semaglutide; 39·0–84·6% of participants who received elecoglipron had an HbA_{1c} of 6·5% or less versus 12·5% in the placebo group and 54·6% in the oral semaglutide group at week 26 (figure 3; appendix 1 p 14).

All doses of elecoglipron were superior to placebo for mean change from baseline in fasting glucose at week 26, with –1·19 mmol/L for participants in the 5 mg group, –1·74 mmol/L for those in the 15 mg group, –1·92 mmol/L for those in the 25 mg group, –1·85 mmol/L for those in the 50 mg group, –2·41 mmol/L for those in the 75 mg (every 2-week escalation) group, and –2·86 mmol/L for those in the 75 mg (every 4-week escalation) group versus 0·02 mmol/L for those in the placebo group and –1·68 mmol/L for those in the semaglutide group. Reductions in fasting glucose were observed for all doses as early as 4 weeks of treatment (figure 3).

At week 26, mean absolute change from baseline in bodyweight for participants who received elecoglipron was –2·54 kg for those in the 5 mg group, –3·42 kg for those in the 15 mg group, –4·35 mg for those in the 25 mg group, –5·79 mg for those in the 50 mg group, –7·61 kg for those in the 75 mg (every 2-week escalation) group, and –6·52 kg for those in the 75 mg (every 4-week escalation) group versus –1·94 kg for those in the placebo group and –4·69 kg for those in the semaglutide 14 mg group (figure 3). Elecoglipron doses of 25 mg and higher were superior to placebo with respect to mean absolute change from baseline in bodyweight of –2·40 to –5·67 kg. Bodyweight continued to decrease up to week 26 (figure 3). At week 26, the mean percent change from baseline in bodyweight for participants who received elecoglipron was –2·9% for those in the 5 mg group, –3·4% for those in the 15 mg group, –4·0% for those in the 25 mg group, –5·9% for those in the 50 mg group, –7·7% for those in the 75 mg group (every 2-week escalation), –7·0% for those in the 75 mg group (every 4-week escalation) versus –1·7% for those in the placebo group and –5·1% for those in the semaglutide 14 mg group. Elecoglipron doses of 25 mg and greater were superior to placebo with respect to mean percent change from baseline in bodyweight between-group treatment differences of –2·3% to –6·1%.

The estimated proportion of participants on elecoglipron achieving weight reductions of at least 5% from baseline at week 26 was greater with elecoglipron 15 mg and above (34·4–72·3%) than with placebo (20·2%). 45·4% of participants on oral semaglutide achieved weight reduction of at least 5% from baseline at week 26. 4·4–32·7% of participants treated with elecoglipron achieved at least a 10% reduction in their bodyweight at week 26 versus 4·5% in the placebo group and 24·0% of participants in the oral semaglutide group (figure 3; appendix 1 p 14). Results of primary and key secondary efficacy endpoints for the supplementary estimand and pre-specified subgroup analyses for the primary endpoint are presented in appendix 1 (pp 16, 20). In pre-specified subgroup analyses, greater reductions in HbA_{1c} were observed among participants with higher baseline HbA_{1c} values (>8%) versus those with lower values; treatment effects on HbA_{1c} appeared consistent regardless of sex, BMI, or background type 2 diabetes medication (appendix 1 p 20).

One or more adverse events ranged from 63–87% of elecoglipron-treated participants, compared with 63% of placebo-treated participants and 67% of oral semaglutide treated participants. Most adverse events were gastrointestinal, reported by 32–71% of participants treated with elecoglipron, compared with 27% treated with placebo and 31% treated with semaglutide. Gastrointestinal adverse events occurred more frequently in participants in the 25 mg fixed-dose treatment group (63%; 26 of 41) and in the 75 mg every 4-week dose-escalation treatment group (71%; 27 of 38). Gastrointestinal adverse events were generally dose-dependent and the majority were mild to moderate in severity, with few events leading to permanent discontinuation (five or fewer events in all elecoglipron groups, three in the semaglutide group, and two in the placebo group; table 3). Nausea, constipation, diarrhoea, and vomiting were the most common adverse events in participants treated with elecoglipron (table 3). Across elecoglipron groups, nausea occurred in 3% (one of 38) to 37% (14 of 38) of participants versus 3% (two of 71) with placebo and vomiting occurred in 3% (one of 38) to 18% (seven of 38) versus 1% (one of 71) with placebo. Rates of nausea and vomiting with open-label semaglutide were 18% (nine of 51) and 4% (two of 51), respectively. Adverse event count and rate are provided in appendix 1 (p 19). Serious adverse events ranged from 0% to 10% in elecoglipron-treated participants, 8% in placebo-treated participants, and 2% in semaglutide-treated participants. One participant in the placebo group died due to a haemorrhagic stroke during the trial (table 3).

Discontinuation of trial treatment related to an adverse event occurred most frequently in the elecoglipron 25 mg fixed-dose group (17%; seven of 41 participants) and in the elecoglipron 75 mg every 4-week escalation group (16%; six of 38 participants; table 3).

Three participants (1%; three of 282) who received elecoglipron (in the 15 mg, 50 mg, and 75 mg every 2-week escalation groups) reported adverse events of hypoglycaemia or blood glucose decreased. Cases were level 1 hypoglycaemia (50 mg and 75 mg) or level 2 hypoglycaemia (15 mg). There were no serious adverse events or discontinuations due to hypoglycaemia (table 3). Liver transaminases improved in participants receiving elecoglipron compared with with those receiving placebo (appendix 1 p 18).

Discussion

In this phase 2b dose-ranging trial in participants with type 2 diabetes, treatment with daily oral elecoglipron resulted in glucose lowering and reduction in bodyweight from baseline to week 26. At all dose concentrations of elecoglipron, reductions in HbA_{1c} and fasting glucose at week 26 were larger than with placebo, with a high proportion reaching an HbA_{1c} of less than 7% (up to 89·6%) and HbA_{1c} of 6·5% or less (up to 84·6%) compared with placebo (24·9% and 12·5%, respectively). Weight reductions increasing with dose were observed at week 26 in participants treated with elecoglipron. Up to 72·3% achieved at least 5% weight reduction from baseline at week 26 compared with 20·2% with placebo. As a benchmark exploratory comparison with the oral GLP-1 receptor agonist, semaglutide, similar or greater glucose-lowering efficacy as well as weight reduction at higher doses of elecoglipron was observed. Elecoglipron also demonstrated a safety and tolerability profile consistent with those reported in other GLP-1 receptor agonist trials at a similar phase of development.[12,13]

Notably, the population in SOLSTICE represents a relatively early type 2 diabetes treatment population, evidenced by a median duration of type 2 diabetes of 6·2 years, with 22% treated with diet and exercise alone at baseline. The high rates of achieving clinical glycaemic and weight targets for this population in the SOLSTICE trial speaks to the potential advantage of efficacious therapies early in the course of type 2 diabetes.[14] Furthermore, younger patients and those with shorter duration of type 2 diabetes typically have greater levels of obesity,[15,16] highlighting the importance of the dual care goals in this population. With the greater recognition of the long-term consequences of hyperglycaemia and obesity, early treatment approaches that effectively target both glycaemia and bodyweight can, in part, contribute to long-term benefits, as they have the potential to shift the longitudinal trajectory and risk of multiple long-term conditions that accompany these risk factors.[17] International guidelines support earlier use of therapies that address a comprehensive approach to diabetes care, including reductions in HbA_{1c} and bodyweight in order to decrease risk of long-term complications. Hence, early considerations in type 2 diabetes care are increasingly shifting towards combined treatment goals and GLP-1 receptor agonists are well suited to support these objectives.

Development of next-generation oral small molecule GLP-1 receptor agonists might enhance patient care and support patient preferences by eliminating administration restrictions and by removing any existing barriers related to injectable delivery. Further advancement of the oral small molecule GLP-1 receptor agonist class might also open the possibility for future combination therapies with other oral therapy to support longitudinal care goals.

Our findings in the SOLSTICE trial also highlight the intricate and multifactorial dosing considerations unique to GLP-1 receptor agonist development. Overall, elecoglipron was safe and well tolerated, with a profile consistent with the GLP-1 receptor agonist class. Gastrointestinal effects were the main consideration, with gastrointestinal events and treatment discontinuation being generally dose-related and dose-escalation schedule related. As shown in SOLSTICE, there are three factors in considering dose-related tolerability: maximum target dose, with higher gastrointestinal symptoms reported at the highest dose (75 mg); starting dose, with higher gastrointestinal symptoms reported with higher starting doses (eg, 15% vomiting with fixed-dose initiation at 25 mg compared with 3% with 5 mg fixed dose); and pace of dose escalation. These factors will inform and guide phase 3 dose selection, striking a balance between overall tolerability, individual flexibility, and achievable efficacy. Similarly, the oral peptide semaglutide and oral small molecule orforglipron clinical development programmes leveraged essential phase 2 dose–response data to inform these three GLP-1 receptor agonist dosing factors in phase 3 design, resulting in greater tolerability in the respective phase 3 clinical programmes. Moreover, the efficacy and tolerability findings of elecoglipron are thus far comparable with phase 2 findings from both oral semaglutide and orforglipron programmes,[12,13] providing further rationale for advancement to phase 3.

There are several strengths to this trial, including the permutations of dosing considerations and testing of target dose, starting dose, and dose-escalation scheme. Furthermore, the inclusion of an exploratory reference comparator group with oral semaglutide provides a contemporary clinical benchmark by comparing elecoglipron to the current available oral GLP-1 receptor agonist in type 2 diabetes. Additionally, the high 95% trial completion rate is a testament to the participants and sites and their commitment to the trial, lending credence to the findings. Furthermore, pre-specified subgroup analyses for the primary endpoint were performed and were in line with the overall study findings. A greater reduction in HbA_{1c} was observed with higher baseline HbA_{1c} values (>8%) versus placebo; treatment effects on HbA_{1c} appeared similar regardless of sex, BMI, or type 2 diabetes background medication. However, given the small sample size, results should be interpreted with caution.

Limitations of this trial include the small treatment group sample size, which limits generalisability. In this phase 2b trial with eight treatment groups, the type 1 error was not controlled for multiplicity and all p values should be interpreted descriptively. The open-label oral semaglutide group was included as an exploratory comparator, which does not allow for direct comparisons with elecoglipron treatment groups. As this trial was limited to participants treated with diet and exercise or monotherapy, further studies are needed in a broader and more advanced type 2 diabetes population. Furthermore, while standard of care guidance was provided as per protocol, detailed low-calorie or specific dietary guidance was not part of the scope of the trial. Finally, the 26-week duration is relatively short for assessing long-term efficacy, durability, longitudinal cardiometabolic effects, and safety, warranting longer duration of intervention in future studies.

In summary, elecoglipron achieved greater glucose-lowering effects and weight loss at all dose concentrations relative to placebo in the SOLSTICE trial. Overall, the safety and tolerability profile was consistent with the GLP-1 receptor agonist class. Collectively, the findings support further development of elecoglipron in the phase 3 (ELUMINATE) trial programme.

Contributors

VRA, JR, JM, and JLS contributed to conceptualisation. VRA, JM, and JLS contributed to methodology. VLJ and MM contributed to data curation and analysis. VRA, MJD, JR, JA, AC, RJM, JM, JLS, VLJ, MM, OE, DZ, and MS contributed to data interpretation. VRA and MJD verified the underlying data. All authors approved the final version of the manuscript and vouch for data accuracy and fidelity of the trial to the protocol. All authors had full access to all data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

VRA declares institutional contracts from Amgen, Applied Therapeutics, AstraZeneca, Biomea, Boehringer Ingelheim, Corcept, Eli Lilly, Fractyl, Kailera, Novo Nordisk, Pfizer, Recordati, Rhythm, and Servier; and consulting fees from Baim Institute for Clinical Research, Mediflix, Sanofi, and Roche. MJD declares consulting, advisory, or speaker fees from AbbVie, Amgen, AstraZeneca, Biomea Fusion, Boehringer Ingelheim, Carmot Roche, Daewoong Pharmaceutical, EktaH, Eli Lilly, GSK, Innovent Bio, Kailera, Novo Nordisk, Regeneron, Sanofi, Zealand Pharma, and Zuellig Pharma; and institutional research grants from AstraZeneca, Boehringer Ingelheim, and Novo Nordisk. JR declares clinical research grants from Amgen, Applied Therapeutics, AstraZeneca, Biomea Fusion, Boehringer Ingelheim, Corcept, Corxel, Eli Lilly, Hanmi, Merck, Novo Nordisk, Pfizer, Regeneron, Roche, Regor, Sanofi, Structure Therapeutics, and Terns; advisory or consulting fees from Amgen, Applied Therapeutics, Arrowhead, Biomea Fusion, Boehringer Ingelheim, Corcept, Eccogene, Eli Lilly, Hanmi, i2O Therapeutics, Novo Nordisk, Regeneron, Roche, Sanofi, Structure Therapeutics, and Terns; and has received honoraria for lectures from Eli Lilly, Novo Nordisk, and Sanofi. JA has acted as a consultant and advisor, clinical investigator, and speaker for Eli Lilly, Novo Nordisk, AstraZeneca, Boehringer Ingelheim, and Amgen. AC has received speaking fees from and is a member of advisory boards for AstraZeneca, Boehringer-Ingelheim, Eli-Lilly, Novo Nordisk, Sanofi, and Menarini and has received research grants from Eli Lilly, Novo Nordisk, and Menarini. AC is also a member of the data monitoring committee of Boehringer Ingelheim. RJM declares advisory, consulting, and speaker fees from Sanofi, NovoNordisk, and Eli Lilly; and institutional research grants from AstraZeneca, Boerhinger-Ingleheim, Eli Lilly, NovoNordisk, and Roche. JA reports consulting fees or speaking honoraria from AstraZeneca, Boehringer Ingelheim, Lilly, and Novo Nordisk. JM, MM, VLJ, OE, JLS, MS, and DZ are employees and stockholders of AstraZeneca.

Data sharing

Data underlying the findings described in this manuscript may be obtained in accordance with AstraZeneca's data sharing policy. Data for studies directly listed on Vivli can be requested through Vivli. Data for studies not listed on Vivli could be requested through Vivli. AstraZeneca Vivli member page is also available outlining further details.

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